Omni-WorldBench: Towards a Comprehensive Interaction-Centric Evaluation for World Models

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Research Questions

- How can we systematically evaluate the interactive response capabilities of video-based world models across different interaction complexity levels?
- What are the current limitations of state-of-the-art world models in handling causally consistent action-driven state transitions?
- How do different world model paradigms (T2V, IT2V, camera-controlled) compare in their ability to maintain temporal coherence and physical plausibility under interactive conditions?

Core Idea

Existing world model benchmarks focus on visual quality metrics (FID, FVD) or static 3D consistency, missing the core capability of world models: interactive response. The key insight is that true world modeling requires evaluating how actions causally drive state transitions across space and time, not just visual fidelity. The paper introduces a hierarchical interaction framework (Level 1: self-contained actions, Level 2: object-to-object interactions, Level 3: global environmental changes) combined with an agent-based evaluation system that adaptively weights different evaluation dimensions based on prompt semantics.

Key Formula

InterStab-L Similarity

$$s(i,j) = \frac{1}{2}\left(\mathrm{SSIM}_{\mathrm{gray}}(I_i, I_j) + \cos(\phi(I_i), \phi(I_j))\right)$$

Composite similarity metric for long-horizon temporal coherence. I_i, I_j are frame pairs at revisit timestamps, SSIM_gray measures structural similarity, φ(·) is CLIP visual encoder for semantic consistency. Combines low-level structural fidelity with high-level semantic consistency to assess temporal stability.

AgenticScore

$$\mathrm{AgenticScore} = w_1 A_I + w_2 A_G + w_3 A_C$$

Final adaptive evaluation score where A_I is interaction effect fidelity, A_G is generated video quality, A_C is camera-object controllability, and w_1, w_2, w_3 are weights determined by MLLM analysis of prompt semantics. Enables context-aware aggregation of heterogeneous evaluation dimensions.

Key Figure

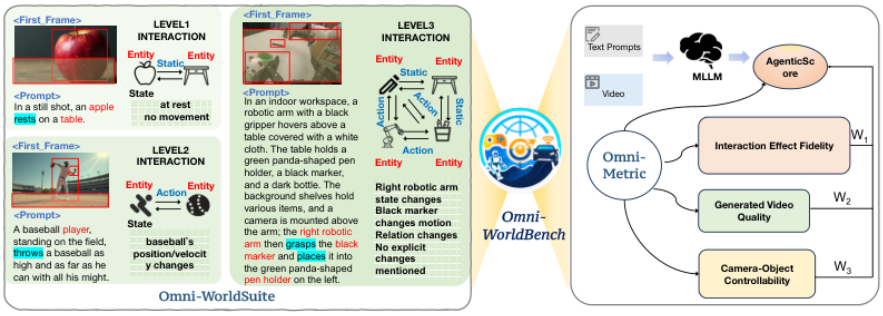

Figure 1: Overview of Omni-WorldBench. **Left:** Omni-WorldSuite defines three levels of interactions, each specified by an initial frame and a prompt. **Right:** Omni-Metrics comprises an evaluation pipeline that measures interaction effect fidelity, generated video quality, camera-object controllability, and spatiotemporal causal coherence, and then employs an MLLM to adaptively fuse these signals into the final AgenticScore.

Figure 1: Shows the complete Omni-WorldBench framework with three-level interaction hierarchy examples and the multi-dimensional evaluation pipeline leading to AgenticScore.

Limitations

- Cannot fully capture complexity of open-world interactive environments, especially long-horizon and highly dynamic settings
- Limited to three interaction levels which may not cover all possible interaction complexities in real-world scenarios
- Human-aligned evaluation results planned for future release, current validation relies primarily on automatic metrics
- Computational cost of comprehensive evaluation across 15 metrics may limit scalability for frequent model assessment